Fine-tuning AlphaGenome on single-cell pseudobulk multiome data

1 Summary

One AlphaGenome [1] model was fine-tuned jointly on five single-cell multiome studies (ten species-specific sources, 20 output heads). It starts from the AlphaGenome FOLD_1 checkpoint and is evaluated on genome regions that this checkpoint never saw as pretraining targets. The most effective changes were to the training targets: RNA rebuilt as gene-level pseudobulk counts per million (CPM) from raw counts, ATAC rebuilt from raw fragments where released tracks were unsuitable, and removal of near-empty, unsupported or artifactual signal. On held-out regions, frozen-backbone heads reach mean correlation $R = 0.455$, and adding low-rank adapters raises it to 0.500–0.507 (ATAC 0.66–0.67, RNA 0.33–0.35). LoRA and LoRA with convolutional adapters (LoCon) perform similarly. On fine-mapped brain eQTLs, gene-span variant effects give AUROC up to 0.57 (SingleBrain) and 0.64 (XQTL). The frozen-backbone model is best for eQTLs, adapter training lowers eQTL discrimination, and discrimination is concentrated within 10 kb of the TSS.

An earlier version of this report fine-tuned from the `all_folds` checkpoint and held out chromosomes 8 and 9, which that checkpoint was pretrained on. Its held-out results (mean test R 0.627) are superseded.

2 Data

Each study provides pseudobulk ATAC and RNA profiles for annotated cell groups (Table 1) [3, 4, 5, 6, 7]. Each source keeps its own genome assembly. Macaque and marmoset sequences use AlphaGenome's human organism embedding.

Table 1: Training data. Group counts are ATAC/RNA cell groups; Liu and Johansen RNA heads have one channel per strand. ATAC source is the released track or a reconstruction from raw fragments.

Study	Tissue	Species	Groups	ATAC source
Mannens (HDA) [3]	first-trimester brain	human	134/134	released
Liu (HDMA) [4]	fetal, 12 organs	human	186/186	fragments
Johansen 2025 [5]	basal ganglia	human, macaque, marmoset	47/47	fragments
Zemke 2023 [6]	motor cortex	human, macaque, marmoset, mouse	20/19	released
Zemke 2024 [7]	hippocampus (aging)	human	18/18	fragments

Held-out regions. AlphaGenome's FOLD_1 checkpoint [1] was pretrained without two of the eight Borzoi genome folds [2], one used here for validation and one for test. Input windows of 131,072 bp are assigned to training, validation or test by these folds, directly for human and mouse and through liftover of 4 kb bins to hg38 for macaque and marmoset, so that sequence orthologous

to held-out human regions is also held out. Windows that overlap more than one split are dropped (0.4–4.4% per source), and validation and test windows are subsampled to about 1,000 per source.

3 Training strategies

Performance is measured by signed double-centered Pearson correlation R. For target and prediction matrices $Y, \hat{Y} \in \mathbb{R}^{N \times C}$, with N genomic elements (genes for RNA, 128 bp bins for ATAC) and C cell groups, row and column means are removed before computing the correlation, so R measures relative structure across elements and cell groups rather than assay scale. The comparisons below come from matched pairs of earlier runs that differed in one setting; those runs used the chromosome split, so their absolute values are inflated but the differences between arms are informative.

RNA targets and loss. Released coordinate RNA tracks were replaced with gene-level targets. Raw counts are summed over the cells of each group before CPM normalization; summing per-cell normalized expression instead gives targets whose double-centered agreement with the correct ones is only 0.46–0.54 (Johansen). The model's RNA output $\hat{x} \in \mathbb{R}^{B \times C}$ over B 128 bp bins is read out per gene g as

$$\hat{y}_{g,c} = \sum_b w_{g,b}\, \hat{x}_{b,c}, \tag{1}$$

where $w_{g,b} \in [0, 1]$ is the fraction of bin b covered by exons of g, and the loss is the squared error between $\log(1 + \hat{y}_{g,c})$ and $\log(1 + y_{g,c})$. In a matched frozen-backbone comparison on Zemke 2023 that changed only this target, test RNA R rose from 0.152 to 0.421 with ATAC unchanged. The gene-level target also avoids a ribosomal-repeat artifact in the released marmoset RNA tracks. Adding a per-bin coverage loss did not help (HDA, mean validation R 0.604 with versus 0.598 without), so it was dropped.

ATAC targets. Where released tracks were significance scores or poorly matched coverage, ATAC was rebuilt from raw fragments as full-depth fragment coverage per million fragments, without downsampling. For Johansen human, test R was 0.210 with released tracks and 0.458 with the rebuilt target under the same training setup.

Quality control. One Johansen group with 33,424 fragments (median 1.1×10^8) contributed 48% of all squared ATAC signal and was removed. Groups without supporting cell assignments are masked rather than given zero targets: four Zemke 2024 subtype channels and the Zemke 2023 L5 IT RNA group. The released Zemke 2023 ATAC tracks contain bins with signal of 3,000 to 60,000 RPKM in every cell group, mostly poly-G/C runs, against genome-wide 99.99th percentiles of 97 to 169. In the macaque and marmoset held-out sets a single such bin or window carried 51–89% of the double-centered variance, so validation and test R for these heads measured one or two loci. Following the idea of the ENCODE blacklist [10], a blacklist of bins whose minimum across cell groups exceeds ten times the 99.99th percentile, and of windows whose summed signal exceeds ten times the median window, flags 9 to 19 regions per species. Excluding them removes one held-out window per affected split and leaves no held-out window with more than 4% of the variance. The blacklist is applied at evaluation; the training windows still contain these regions.

Loss. A double-centered correlation term, $1 - R$ computed within each batch, was added to the loss of every RNA head (weight 1, HDA 0.1) and of the Zemke 2023 ATAC heads (weight 3). In a

matched continuation it raised validation R from 0.568 to 0.583, mainly through RNA (0.461 to 0.491).

Model and optimization. The pretrained backbone is shared by all heads. Each source has its own ATAC and RNA heads, because target distributions differ between species of the same study; with equal sampling of sources this raised LoRA test R from 0.584 to 0.605 relative to one pooled head per study and modality, a comparison that changes both factors together. Randomly initialized heads are first trained on a frozen backbone; the best warmup epoch then starts adapter training. Adapters are LoRA [8] (rank 16) on attention and pair projections, optionally with LoCon [9] (rank 4) on downsampling blocks 2–5, at learning rate 3×10^{-4}; learning rate 10^{-3} was unstable for LoCon. Initializing heads from pretrained neural accessibility tracks improved early warmup validation R (0.507 against 0.488 at epoch 4) but was not used in the final runs.

Not helpful. Higher adapter rank (LoRA 32, LoCon 8), uniform RNA loss upweighting, target value transforms, refitting heads after adapter training, and gene-balanced batch ordering did not improve held-out R.

4 Held-out results

The frozen-backbone warmup selected epoch 5 by validation R. Both adapter arms were trained from it to epoch 18 and selected epoch 18 (LoRA) and epoch 15 (LoRA+LoCon). Table 2 lists test R per source with the Zemke 2023 ATAC blacklist applied. Adapters raise mean test R from 0.455 to 0.507 (LoRA) and 0.500 (LoRA+LoCon). LoRA+LoCon is slightly better for ATAC (0.667 against 0.661) and worse for RNA (0.333 against 0.353), so the two adapter strategies are effectively tied. RNA remains the weaker modality and varies between sources, from about 0.2 (Johansen marmoset, HDA with LoCon, Liu) to 0.46 (Zemke 2023 marmoset). Split-half reliability gives ATAC correlation ceilings of 0.82–0.93 for Johansen and 0.90 for Zemke 2024, and RNA ceilings of 0.93 or higher where measured, so RNA is far below its noise limit.

Table 2: Test R on FOLD_1 held-out regions with the Zemke 2023 ATAC blacklist applied.

		Head-only		LoRA		LoRA+LoCon	
Study	Species	ATAC	RNA	ATAC	RNA	ATAC	RNA
HDA	human	0.750	0.152	0.769	0.412	0.771	0.193
Liu	human	0.745	0.218	0.752	0.235	0.750	0.293
Johansen 2025	human	0.559	0.316	0.588	0.426	0.591	0.405
	macaque	0.571	0.282	0.607	0.367	0.613	0.364
	marmoset	0.483	0.232	0.541	0.229	0.552	0.223
Zemke 2023	human	0.656	0.273	0.682	0.319	0.689	0.310
	macaque	0.587	0.320	0.629	0.398	0.638	0.372
	marmoset	0.589	0.423	0.641	0.434	0.652	0.464
	mouse	0.647	0.271	0.687	0.347	0.701	0.372
Zemke 2024	human	0.685	0.333	0.710	0.362	0.715	0.336
Mean		0.627	0.282	0.661	0.353	0.667	0.333

5 Fine-mapped eQTL benchmark

Positives are variants with SuSiE [11] posterior inclusion probability (PIP) above 0.75 and negatives have PIP below 0.01, within 512 kb of the gene's TSS. SingleBrain has 35 fine-mapping panels in seven brain cell classes (54,193 variants), with summary statistics labeled as mmQTL [12] analyses; XQTL has six cell-type panels (10,452 variants). Each variant is scored with the human Zemke 2023 RNA head on a 1 Mb window centered at the TSS. The gene-span effect is the absolute alternate-minus-reference difference in mean prediction over the gene's bins, averaged over the tracks matched to the panel's cell type. The TSS-bin summary performs worse and is not shown.

Table 3 compares the three models with one selected pretrained AlphaGenome track per cell type. The frozen-backbone model is best on both datasets, and adapter training lowers pooled AUROC by 0.01–0.06. The native tracks were selected on the same variants, so their values are optimistic. Discrimination is concentrated near the TSS: AUROC is 0.65–0.69 (SingleBrain) and 0.83–0.88 (XQTL) within 1 kb and falls below 0.5 for all models beyond about 50 kb (SingleBrain) and 100 kb (XQTL) (Figure 1).

Table 3: Gene-span eQTL discrimination. AUROC and AP are pooled over panels; "within" is the mean of per-cell-class AUROC.

	SingleBrain			XQTL		
Model	AUROC	AP	Within	AUROC	AP	Within
Head-only	0.566	0.472	0.559	0.644	0.636	0.648
LoRA	0.511	0.407	0.501	0.614	0.591	0.597
LoRA+LoCon	0.532	0.433	0.527	0.631	0.608	0.624
Native matched tracks	0.524	0.439	0.535	0.627	0.626	0.649

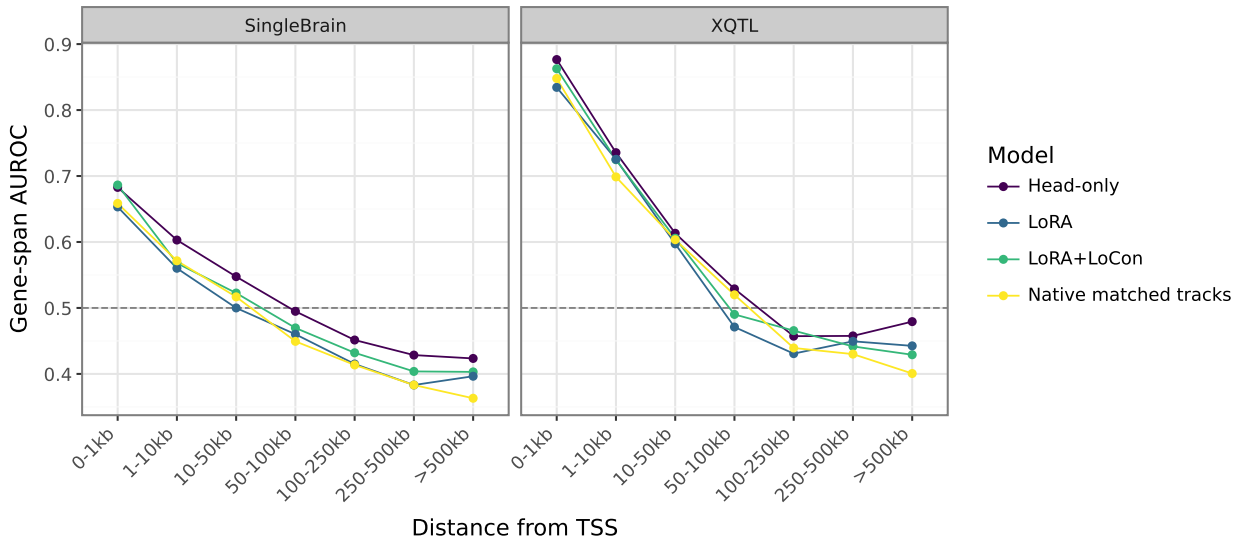


Figure 1: Gene-span AUROC by distance between variant and TSS. The dashed line is chance.

The trained tracks carry a small amount of cell-type-specific signal. Scoring each panel with every cell class's tracks, the matched class exceeds the unmatched ones by 0.012–0.028 AUROC on SingleBrain and 0.039–0.044 on XQTL, which is significant under permutation of the class

assignment for every model ($p \leq 0.006$). The matched composite ranks first for microglia panels in every model and dataset and for most astrocyte and oligodendrocyte panels, whereas excitatory and inhibitory neuron panels show little specificity. Predicted effects are largely shared across cell types, with median Spearman correlation 0.61–0.72 between tracks.

References

[1] Avsec Ž, et al. Advancing regulatory variant effect prediction with AlphaGenome. *Nature* 649, 1206–1218 (2026). doi:10.1038/s41586-025-10014-0

[2] Linder J, et al. Predicting RNA-seq coverage from DNA sequence as a unifying model of gene regulation. *Nat Genet* 57, 949–961 (2025). doi:10.1038/s41588-024-02053-6

[3] Mannens CCA, et al. Chromatin accessibility during human first-trimester neurodevelopment. *Nature* 647, 179–186 (2024). doi:10.1038/s41586-024-07234-1

[4] Liu BB, et al. Multiomics and deep learning dissect regulatory syntax in human development. *Nature* 653, 1240–1253 (2026). doi:10.1038/s41586-026-10326-9

[5] Johansen NJ, et al. Cross-species consensus atlas of the primate basal ganglia. *bioRxiv* (2025). doi:10.64898/2025.12.15.694496

[6] Zemke NR, et al. Conserved and divergent gene regulatory programs of the mammalian neocortex. *Nature* 624, 390–402 (2023). doi:10.1038/s41586-023-06819-6

[7] Zemke NR, et al. Epigenetic and 3D genome reprogramming during the aging of the human hippocampus. *Science* 393, eadt8307 (2026). doi:10.1126/science.adt8307

[8] Hu EJ, et al. LoRA: Low-rank adaptation of large language models. *arXiv* 2106.09685 (2021). arXiv:2106.09685

[9] Yeh S-Y, et al. Navigating text-to-image customization: from LyCORIS fine-tuning to model evaluation. *International Conference on Learning Representations* (2024). arXiv:2309.14859

[10] Amemiya HM, Kundaje A, Boyle AP. The ENCODE blacklist: identification of problematic regions of the genome. *Sci Rep* 9, 9354 (2019). doi:10.1038/s41598-019-45839-z

[11] Wang G, Sarkar A, Carbonetto P, Stephens M. A simple new approach to variable selection in regression, with application to genetic fine mapping. *J R Stat Soc Series B Stat Methodol* 82, 1273–1300 (2020). doi:10.1111/rssb.12388

[12] Zeng B, et al. Multi-ancestry eQTL meta-analysis of human brain identifies candidate causal variants for brain-related traits. *Nat Genet* 54, 161–169 (2022). doi:10.1038/s41588-021-00987-9